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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Microcontroller

Lab Report

Lab 1: LED Animations

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1 Exercise 1

Report 1: Schematic

The Proteus schematic for this exercise is shown below.

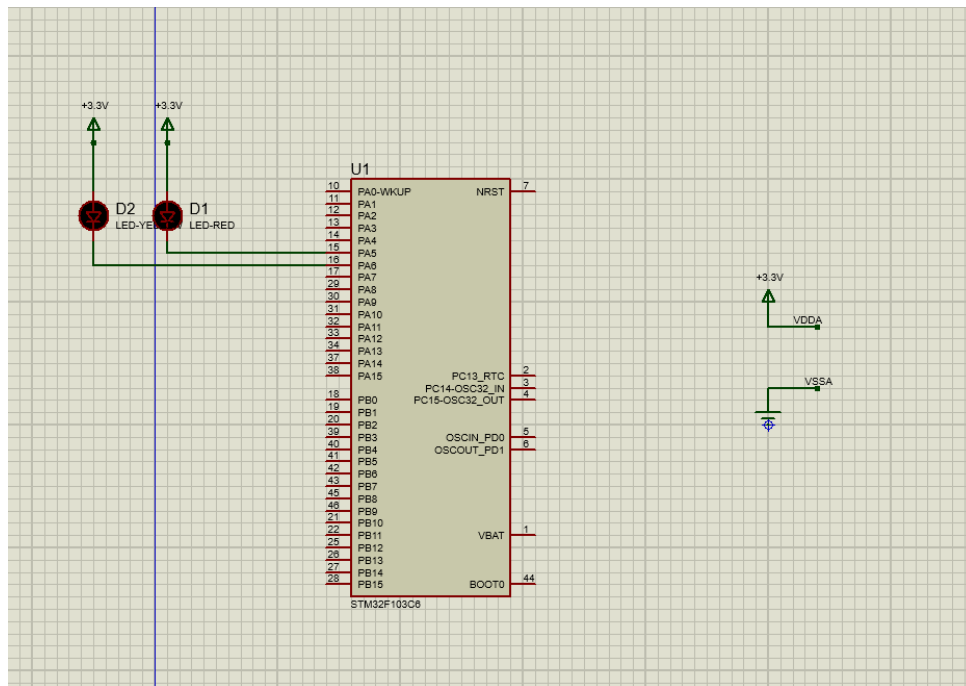


Figure 1.1: https://github.com/aulequangtri/microprocessor_lab

Report 2: Source Code

The source code inside the `while(1)` loop to switch the states of the RED and YELLOW LEDs every 2 seconds:

```
1 while (1)
2 {
3     HAL_GPIO_TogglePin(LED_RED_GPIO_Port, LED_RED_Pin);
4     HAL_GPIO_TogglePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin);
5     HAL_Delay(1000);
6 }
```

2 Exercise 2

Report 1: Schematic

The Proteus schematic for the 3-LED traffic light system is shown below.

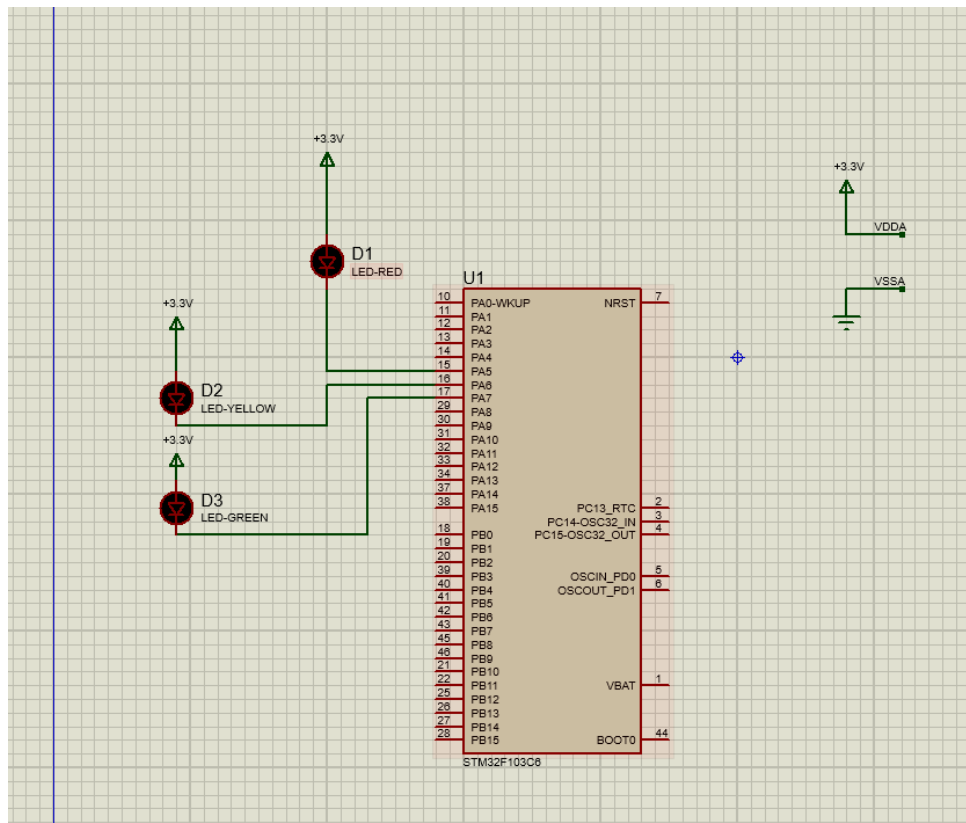


Figure 2.1: Schematic for Exercise 2 (Traffic Light)

Report 2: Source Code

The source code inside the `while(1)` loop simulating the traffic light cycle:

```
1 while (1)
2 {
3     HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin, RESET);
4     HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin, SET);
5     HAL_GPIO_WritePin(LED_GREEN_GPIO_Port, LED_GREEN_Pin, SET);
6     HAL_Delay(5000);
7 }
```



```
8     HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin, SET);
9     HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin, SET);
10    HAL_GPIO_WritePin(LED_GREEN_GPIO_Port, LED_GREEN_Pin, RESET);
11    HAL_Delay(3000);
12
13    HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin, SET);
14    HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin, RESET);
15    HAL_GPIO_WritePin(LED_GREEN_GPIO_Port, LED_GREEN_Pin, SET);
16    HAL_Delay(2000);
17 }
```

3 Exercise 4

Report 1: Schematic

The schematic for the 7-segment display integration.

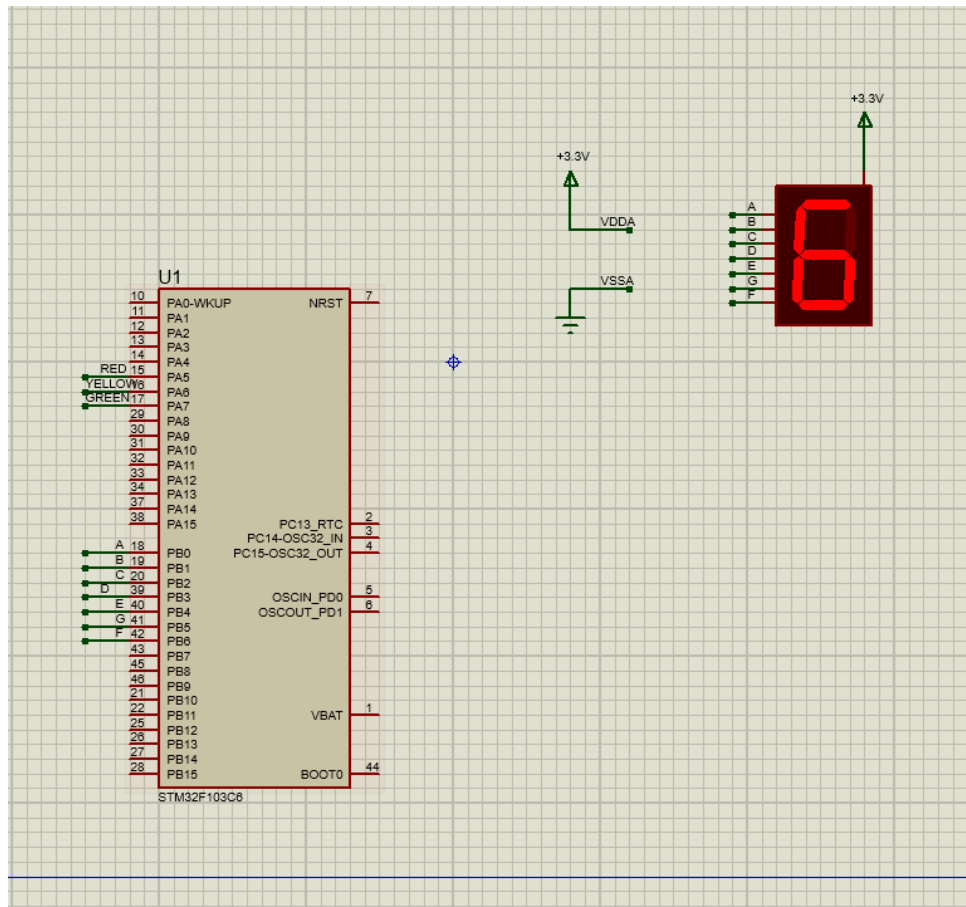


Figure 3.1: Schematic for Exercise 4 (7-Segment LED)

Report 2: Source Code for display7SEG function

The function `display7SEG(int num)` is implemented as follows:

```
1 void display7SEG(int num) {  
2     /* Turn off all segments first */  
3     HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin, GPIO_PIN_SET);  
4     HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin, GPIO_PIN_SET);  
5     HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin, GPIO_PIN_SET);  
6     HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin, GPIO_PIN_SET);  
7     HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin, GPIO_PIN_SET);  
8     HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin, GPIO_PIN_SET);  
9     HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin, GPIO_PIN_SET);  
10 }
```



```
11      /* Turn on required segments based on the number */
12      switch (num) {
13          case 0:
14              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
15                               GPIO_PIN_RESET);
16              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
17                               GPIO_PIN_RESET);
18              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
19                               GPIO_PIN_RESET);
20              HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
21                               GPIO_PIN_RESET);
22              HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin,
23                               GPIO_PIN_RESET);
24              HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,
25                               GPIO_PIN_RESET);
26              break;
27          case 1:
28              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
29                               GPIO_PIN_RESET);
30              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
31                               GPIO_PIN_RESET);
32              break;
33          case 2:
34              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
35                               GPIO_PIN_RESET);
36              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
37                               GPIO_PIN_RESET);
38              HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
39                               GPIO_PIN_RESET);
40              HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin,
41                               GPIO_PIN_RESET);
42              HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,
43                               GPIO_PIN_RESET);
44              break;
45          case 3:
```

```
33     HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,  
    GPIO_PIN_RESET);  
34     HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,  
    GPIO_PIN_RESET);  
35     HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,  
    GPIO_PIN_RESET);  
36     HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,  
    GPIO_PIN_RESET);  
37     HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,  
    GPIO_PIN_RESET);  
38     break;  
39 case 4:  
40     HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,  
    GPIO_PIN_RESET);  
41     HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,  
    GPIO_PIN_RESET);  
42     HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,  
    GPIO_PIN_RESET);  
43     HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,  
    GPIO_PIN_RESET);  
44     break;  
45 case 5:  
46     HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,  
    GPIO_PIN_RESET);  
47     HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,  
    GPIO_PIN_RESET);  
48     HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,  
    GPIO_PIN_RESET);  
49     HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,  
    GPIO_PIN_RESET);  
50     HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,  
    GPIO_PIN_RESET);  
51     break;  
52 case 6:
```



```
53     HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,  
    GPIO_PIN_RESET);  
54     HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,  
    GPIO_PIN_RESET);  
55     HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,  
    GPIO_PIN_RESET);  
56     HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin,  
    GPIO_PIN_RESET);  
57     HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,  
    GPIO_PIN_RESET);  
58     HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,  
    GPIO_PIN_RESET);  
59     break;  
60     case 7:  
61         HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,  
        GPIO_PIN_RESET);  
62         HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,  
        GPIO_PIN_RESET);  
63         HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,  
        GPIO_PIN_RESET);  
64         break;  
65     case 8:  
66         HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,  
        GPIO_PIN_RESET);  
67         HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,  
        GPIO_PIN_RESET);  
68         HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,  
        GPIO_PIN_RESET);  
69         HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,  
        GPIO_PIN_RESET);  
70         HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin,  
        GPIO_PIN_RESET);  
71         HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,  
        GPIO_PIN_RESET);
```



```
72     HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,  
GPIO_PIN_RESET);  
73     break;  
74     case 9:  
75         HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,  
GPIO_PIN_RESET);  
76         HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,  
GPIO_PIN_RESET);  
77         HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,  
GPIO_PIN_RESET);  
78         HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,  
GPIO_PIN_RESET);  
79         HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,  
GPIO_PIN_RESET);  
80         HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,  
GPIO_PIN_RESET);  
81         break;  
82     }  
83 }
```

4 Exercise 5

Report 1: Schematic

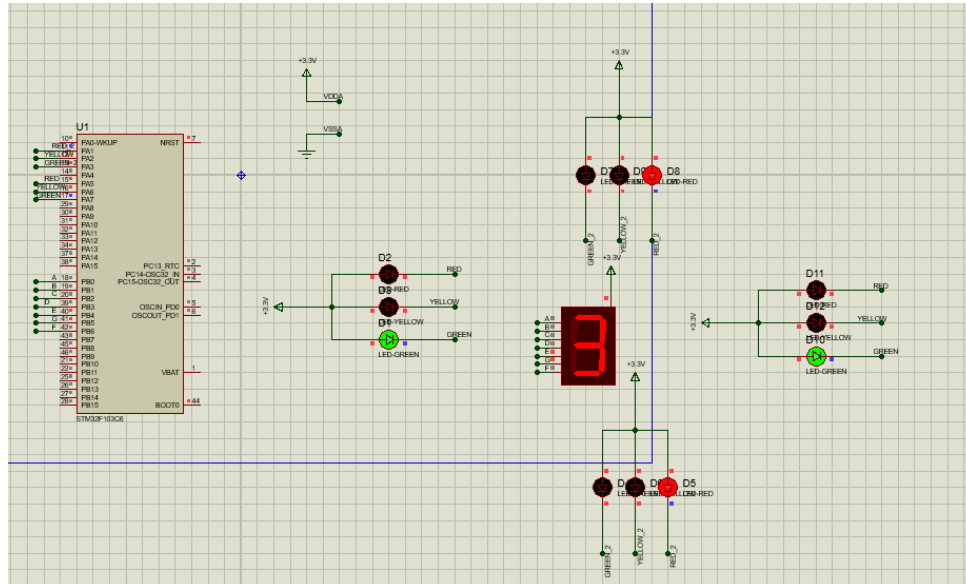


Figure 4.1: Schematic for Exercise 5 (4-Way Traffic Light + 7SEG)

Report 2: Source Code

The code snippet inside the infinite loop for the 4-way traffic light integrated with a 7-segment countdown:

```
1 while (1)
2 {
3     display7SEG(countdown);
4
5     HAL_GPIO_WritePin(RED_GPIO_Port, RED_Pin, GPIO_PIN_SET);
6     HAL_GPIO_WritePin(YELLOW_GPIO_Port, YELLOW_Pin, GPIO_PIN_SET);
7     HAL_GPIO_WritePin(GREEN_GPIO_Port, GREEN_Pin, GPIO_PIN_SET);
8
9     HAL_GPIO_WritePin(RED_2_GPIO_Port, RED_2_Pin, GPIO_PIN_SET);
10    HAL_GPIO_WritePin(YELLOW_2_GPIO_Port, YELLOW_2_Pin, GPIO_PIN_SET);
11    HAL_GPIO_WritePin(GREEN_2_GPIO_Port, GREEN_2_Pin, GPIO_PIN_SET);
12
```

```
13  if (state == 0) {
14      HAL_GPIO_WritePin(RED_GPIO_Port, RED_Pin, GPIO_PIN_RESET);
15      HAL_GPIO_WritePin(GREEN_2_GPIO_Port, GREEN_2_Pin,
16                          GPIO_PIN_RESET);
17
18      countdown--;
19      if (countdown <= 0) {
20          state = 1;
21          countdown = 2;
22      }
23  }
24  else if (state == 1) {
25      HAL_GPIO_WritePin(RED_GPIO_Port, RED_Pin, GPIO_PIN_RESET);
26      HAL_GPIO_WritePin(YELLOW_2_GPIO_Port, YELLOW_2_Pin,
27                          GPIO_PIN_RESET);
28
29      countdown--;
30      if (countdown <= 0) {
31          state = 2;
32          countdown = 3;
33      }
34  }
35  else if (state == 2) {
36      HAL_GPIO_WritePin(GREEN_GPIO_Port, GREEN_Pin, GPIO_PIN_RESET);
37      HAL_GPIO_WritePin(RED_2_GPIO_Port, RED_2_Pin, GPIO_PIN_RESET);
38
39      countdown--;
40      if (countdown <= 0) {
41          state = 3;
42          countdown = 2;
43      }
44  }
45  else if (state == 3) {
46      HAL_GPIO_WritePin(YELLOW_GPIO_Port, YELLOW_Pin,
47                          GPIO_PIN_RESET);
```

```
45     HAL_GPIO_WritePin(RED_2_GPIO_Port, RED_2_Pin, GPIO_PIN_RESET);
46
47     countdown--;
48     if (countdown <= 0) {
49         state = 0;
50         countdown = 3;
51     }
52 }
53
54 HAL_Delay(1000);
55 }
```

5 Exercise 6

Report 1: Schematic

The Proteus schematic for the analog clock using 12 LEDs.

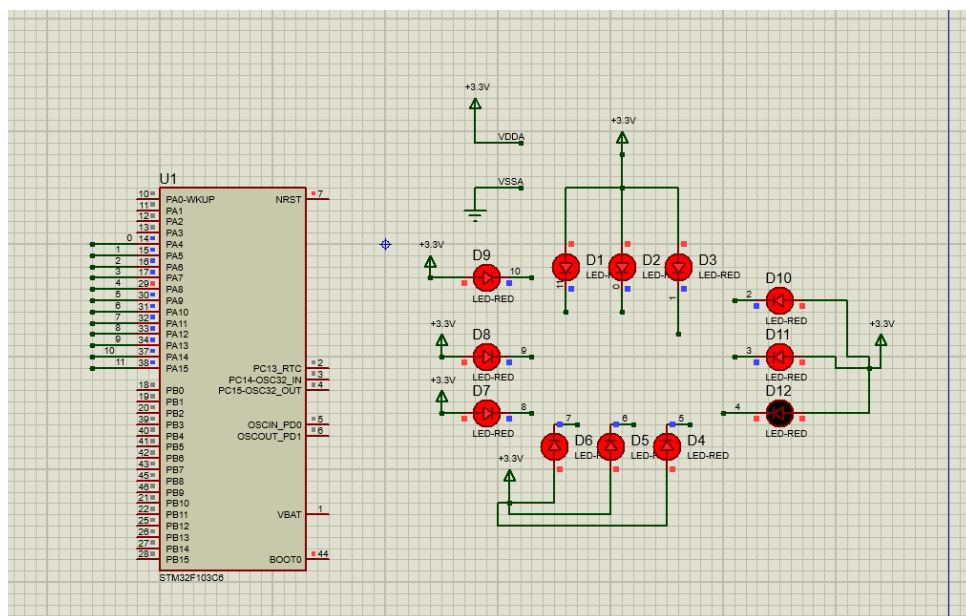


Figure 5.1: Schematic for Exercise 6 (Analog Clock)

Report 2: Source Code

The testing program turning every LED in a sequence:

```
1  while (1)
2  {
3      HAL_GPIO_WritePin(GPIOA, clock_pins[current_led], GPIO_PIN_SET);
4
5      current_led++;
6
7      if (current_led >= 12) {
8          HAL_Delay(1000);
9
10         for (int i = 0; i < 12; i++) {
11             HAL_GPIO_WritePin(GPIOA, clock_pins[i], GPIO_PIN_RESET);
12         }
13         current_led = 0;
14     }
15
16     HAL_Delay(500);
17 }
```

6 Exercise 7 to 10

The following schematic is used for the analog clock exercises (7 to 10):

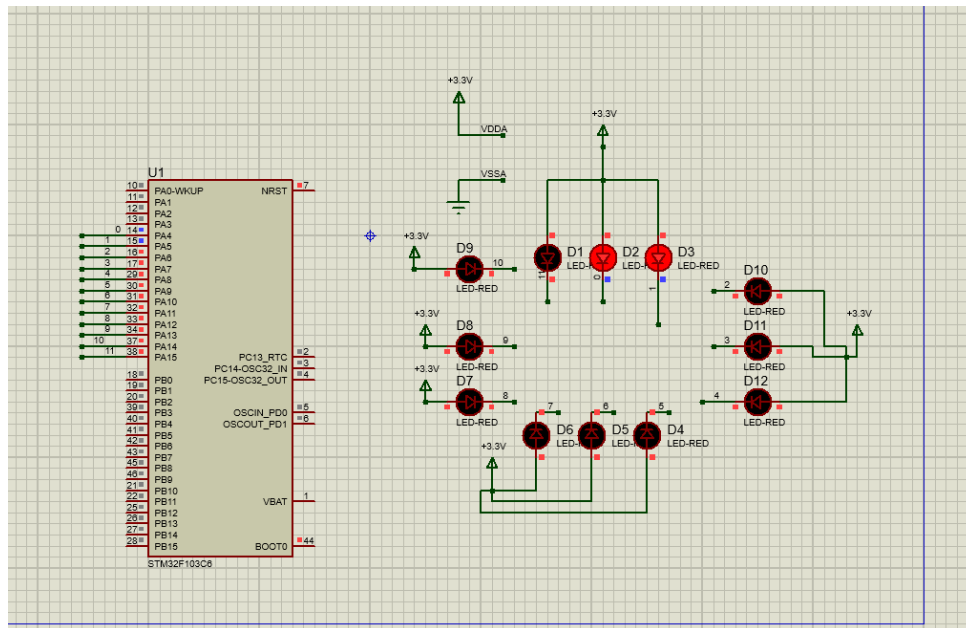


Figure 6.1: Schematic for Exercises 7-10 (Analog Clock Functions)

Exercise 7: clearAllClock()

```

1 void clearAllClock(void) {
2     uint16_t all_clock_pins = GPIO_PIN_4 | GPIO_PIN_5 | GPIO_PIN_6 |
                                GPIO_PIN_7 |
3                                     GPIO_PIN_8 | GPIO_PIN_9 | GPIO_PIN_10 |
                                GPIO_PIN_11 |
4                                     GPIO_PIN_12 | GPIO_PIN_13 | GPIO_PIN_14
                                | GPIO_PIN_15;
5
6     HAL_GPIO_WritePin(GPIOA, all_clock_pins, LED_OFF);
7 }

```

Exercise 8: setNumberOnClock(int num)

```

1 void setNumberOnClock(int num) {
2     if (num >= 0 && num <= 11) {
3         uint16_t clock_pins[12] = {
4             GPIO_PIN_4, GPIO_PIN_5, GPIO_PIN_6, GPIO_PIN_7,

```

```
5         GPIO_PIN_8, GPIO_PIN_9, GPIO_PIN_10, GPIO_PIN_11,  
6         GPIO_PIN_12, GPIO_PIN_13, GPIO_PIN_14, GPIO_PIN_15  
7     };  
8     HAL_GPIO_WritePin(GPIOA, clock_pins[num], LED_ON);  
9 }  
10 }
```

Exercise 9: clearNumberOnClock(int num)

```
1 void clearNumberOnClock(int num) {  
2     if (num >= 0 && num <= 11) {  
3         uint16_t clock_pins[12] = {  
4             GPIO_PIN_4, GPIO_PIN_5, GPIO_PIN_6, GPIO_PIN_7,  
5             GPIO_PIN_8, GPIO_PIN_9, GPIO_PIN_10, GPIO_PIN_11,  
6             GPIO_PIN_12, GPIO_PIN_13, GPIO_PIN_14, GPIO_PIN_15  
7         };  
8         HAL_GPIO_WritePin(GPIOA, clock_pins[num], LED_OFF);  
9     }  
10 }
```

Exercise 10: Analog Clock Main Loop

The integration of the system using 12 LEDs to display an analog clock (Hour, Minute, and Second).

```
1 while (1) {  
2     clearAllClock();  
3  
4     int hour_led = hour % 12;  
5     int min_led = minute / 5;  
6     int sec_led = second / 5;  
7  
8     setNumberOnClock(hour_led);  
9     setNumberOnClock(min_led);  
10    setNumberOnClock(sec_led);
```



```
11
12     HAL_Delay(1000);
13
14     second++;
15     if (second >= 60) {
16         second = 0;
17         minute++;
18
19         if (minute >= 60) {
20             minute = 0;
21             hour++;
22
23             if (hour >= 12) {
24                 hour = 0;
25             }
26         }
27     }
28 }
```